

Homework 8

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Homework

8

ECE 3340 Numerical Methods

Homework 8: Solving Linear Systems

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Problem 1: Loss of Precision

Solve the following linear system using (1) Gaussian Elimination and (2) Partial Pivoting with 3 digits of precision. What is the relative error for each?

$$m_{21} = \frac{2.34}{0.0133} \approx 176$$

$$R_2 \leftarrow R_2 - 176R_1$$

$$176(1.24) \approx 218$$

$$176(1) = 176$$

$$[2.34 \ 1.77 \ 1] - [2.34 \ 218 \ 176] = [0 \ -216 \ -175]$$

$$\begin{bmatrix} 0.0133 & 1.24 & 1 \\ 0 & -216 & -175 \end{bmatrix} \quad x_2 = \frac{-175}{-216} = 0.81$$

$$E_G = 0.222$$

$$0.0133x_1 + 1.24(0.81) = 1$$

$$1.24(0.81) \approx 1.00 \quad x_1 = 0$$

$$A = \begin{bmatrix} 0.0133 & 1.24 \\ 2.34 & 1.77 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

where

$$x = \begin{bmatrix} -0.184 \\ 0.808 \end{bmatrix}$$

$$\begin{bmatrix} 2.34 & 1.77 & 1 \\ 0.0133 & 1.24 & 1 \end{bmatrix}$$

$$m_{21} = \frac{0.0133}{2.34} = 0.00568$$

$$R_2 \leftarrow R_2 - 0.00568R_1$$

$$0.00568(1.77) = 0.0101$$

$$0.00568(1) = 0.00568$$

$$x_2 = \frac{0.994}{1.23} = 0.808$$

$$2.34x_1 + 1.77(0.808) = 1$$

$$1.77(0.808) \approx 1.43$$

$$2.34x_1 = 1 - 1.43 = -0.430$$

$$x_1 = \frac{-0.430}{2.34} = -0.184$$

$$E_P = 0$$

$$E = \frac{|x - \hat{x}|}{|x|} \quad x = \begin{bmatrix} -0.184 \\ 0.808 \end{bmatrix} \quad \hat{x} = \begin{bmatrix} 0 \\ 0.81 \end{bmatrix}$$

Problem 2: Partial Pivoting

Solve the following linear system using partial pivoting and provide the final indexing vector.

$$m_{32} = \frac{2}{4} = 0.5$$

$$R_3 \leftarrow R_3 - 0.5R_2$$

$$[0 \ 2 \ 4 \ 8 \ 10] - [0 \ 2 \ 2 \ 2 \ 2] = [0 \ 0 \ 2 \ 6 \ 8]$$

$$m_{42} = \frac{1}{4} = 0.25$$

$$R_4 \leftarrow R_4 - 0.25R_2$$

$$[0 \ 1 \ 4 \ 6] - [0 \ 1 \ 1 \ 1] = [0 \ 0 \ 3 \ 5]$$

$$\begin{bmatrix} 2 & 3 & 2 & 7 & -20 \\ 0 & 4 & 4 & 8 & -8 \\ 0 & 0 & 2 & 6 & 8 \\ 0 & 0 & 3 & 5 & 5 \end{bmatrix}$$

$$3x_4 = -5 \Rightarrow x_4 = -\frac{5}{3}$$

$$2x_3 + 6x_4 = -8$$

$$2x_3 + 6(-\frac{5}{3}) = -8$$

$$2x_3 - 10 = -8, \quad x_3 = 1$$

$$4x_2 + 4x_3 + 4x_4 = -4$$

$$4x_2 + 4(1) + 4(-\frac{5}{3}) = -4$$

$$4x_2 - \frac{8}{3} = -4 \Rightarrow x_2 = -\frac{1}{3}$$

$$x_1 = -\frac{14}{3}$$

$$x_2 = -\frac{1}{3}$$

$$x_3 = 1$$

$$x_4 = -\frac{5}{3}$$

$$\begin{bmatrix} 0 & 4 & 4 & 4 \\ 2 & 3 & 2 & 7 \\ 0 & 2 & 4 & 8 \\ 0 & 1 & 1 & 4 \end{bmatrix} \quad x = \begin{bmatrix} -4 \\ -20 \\ -10 \\ -6 \end{bmatrix}$$

(1)

$$2x_1 + 3x_2 + 2x_3 + 7x_4 = -20$$

$$2x_1 + 3(-\frac{1}{3}) + 2(1) + 7(-\frac{5}{3}) = -20$$

$$2x_1 - 1 + 2 - \frac{35}{3} = -20$$

$$2x_1 = -\frac{28}{3} \Rightarrow x_1 = -\frac{14}{3}$$

$3x_2 = -4 \Rightarrow x_2 = -\frac{4}{3}$

$$x_1 = -\frac{14}{3}$$

$$x_2 = -\frac{4}{3}$$

$$x_3 = 1$$

$$x_4 = -\frac{5}{3}$$

$$\ell_1 = 2$$

$$\ell_2 = 1$$

$$\ell_3 = 3$$

$$\ell_4 = 4$$